

1 The Problem

Let $n \in \mathbb{N}$ with $n \geq 2$. We define the index set $N = \{1, 2, \dots, n\}$ and two vectors $p = (p_1, p_2, \dots, p_n), q = (q_1, q_2, \dots, q_n)$ with $n \geq 2$ and $p_i, q_i \in \{0, 1\}$ for $i \in N$. For $i, j \in N$ with $i < j$ let

$$\begin{aligned} \text{one}_i^j(p) &= \#\{k \mid i < k < j, p_k = 1\} \\ \overline{\text{one}}_i^j(p) &= \#\{k \mid 1 \leq k < i, p_k = 1\} + \#\{k \mid j < k \leq n, p_k = 1\} \\ \text{zero}_i^j(p) &= \#\{k \mid i < k < j, p_k = 0\} \\ \overline{\text{zero}}_i^j(p) &= \#\{k \mid 1 \leq k < i, p_k = 0\} + \#\{k \mid j < k \leq n, p_k = 0\} \end{aligned}$$

Definition 1. Let the tuple (n, p, q) be as described above. If there exist $i, j \in N$ with $i < j$, $p_i = p_j$ and

$$\text{one}_i^j(p) > \overline{\text{one}}_i^j(p), \quad (1)$$

$$\text{zero}_i^j(p) > \overline{\text{zero}}_i^j(p), \quad (2)$$

then the following transformation, which we call a swap, may be performed for (i, j)

$$\tau_{ij}(p) = (p_1, p_2, \dots, p_{i-1}, 1 - p_i, p_{i+1}, \dots, p_{j-1}, 1 - p_j, p_{j+1}, \dots, p_n).$$

If some q can be reached from p by a finite sequence of m swaps, i.e.

$$(\tau_{i_m j_m} \circ \dots \circ \tau_{i_2 j_2} \circ \tau_{i_1 j_1})(p) = q, \quad (3)$$

we also write $p \sim q$.

Definition 2. Let (n, p, q) be given with $\#\{i \in N \mid p_i = 1\}$ and $\#\{i \in N \mid p_i = 0\}$ odd. If $p \sim q$ holds, we call (n, p, q) solvable.

And the question of course is: given the tuple (n, p, q) , is (n, p, q) solvable?

2 Solution

Proposition 3. Let (n, p, q) be as in Definition 2. Then n is even.

Proof. Since by assumption $\#\{i \in N \mid p_i = 1\}$ and $\#\{i \in N \mid p_i = 0\}$ are odd, $n = \#\{i \in N \mid p_i = 1\} + \#\{i \in N \mid p_i = 0\}$ must be even. $\square$

Proposition 4. Let (n, p, q) be as in Definition 2. Then for every swap of (i, j)

$$j - i \geq \frac{n}{2} + 1, \quad (4)$$

i.e. the number of indices between i and j must be at least $\frac{n}{2}$.

Proof. As a necessary condition for a swap, conditions (1) and (2) give

$$\text{one}_i^j(p) + \text{zero}_i^j(p) \geq \overline{\text{one}}_i^j(p) + \overline{\text{zero}}_i^j(p) + 2.$$

Bringing in the balance equations

$$\begin{aligned} \text{one}_i^j(p) + \text{zero}_i^j(p) + \overline{\text{one}}_i^j(p) + \overline{\text{zero}}_i^j(p) &= n - 2, \\ \text{one}_i^j(p) + \text{zero}_i^j(p) &= j - i - 1 \end{aligned}$$

we obtain

$$\begin{aligned} \text{one}_i^j(p) + \text{zero}_i^j(p) &\geq n - (\text{one}_i^j(p) + \text{zero}_i^j(p)) \\ 2(j-i-1) &\geq n \\ j-i &\geq \frac{n}{2} + 1. \end{aligned}$$

□

Definition 5. Let (n, p, q) be as in Definition 2. We define $L = \{1, \dots, \frac{n}{2} - 1\}$ and $R = \{\frac{n}{2} + 2, \dots, n\}$.

Corollary 6. Let (n, p, q) be as in Definition 2. Then the indices $\{\frac{n}{2}, \frac{n}{2} + 1\}$ can never swap. A swap (i, j) always takes place with an index $i \in L$ and an index $j \in R$.

Definition 7. Let (n, p, q) be as in Definition 2. We call (n, p, q) center-equal if $p_{\frac{n}{2}} = q_{\frac{n}{2}}$ and $p_{\frac{n}{2}+1} = q_{\frac{n}{2}+1}$ holds.

Corollary 8. Let (n, p, q) be as in Definition 2. If (n, p, q) is solvable, then it is center-equal.

Definition 9. Let (n, p, q) be as in Definition 2. We call $i \in L$ swappable if

$$\text{one}_i^n(p) > \overline{\text{one}}_i^n(p) + p_i - p_n \quad \text{and} \quad (5)$$

$$\text{zero}_i^n(p) > \overline{\text{zero}}_i^n(p) - p_i + p_n \quad (6)$$

holds. Analogously, we call $j \in R$ swappable if

$$\text{one}_1^j(p) > \overline{\text{one}}_1^j(p) + p_j - p_1 \quad \text{and} \quad (7)$$

$$\text{zero}_1^j(p) > \overline{\text{zero}}_1^j(p) - p_j + p_1 \quad (8)$$

holds. If all $\{i \in L \mid p_i \neq q_i\}$ and $\{j \in R \mid p_j \neq q_j\}$ are swappable, we call (n, p, q) swappable.

This means that (i, n) can swap as soon as p_n takes on the value of p_i , resp. $(1, j)$ can swap as soon as p_1 takes on the value of p_j .

Proposition 10. Let (n, p, q) be as in Definition 2. If $i \in L, j \in R$ can swap, then both are swappable.

Proof. Since (i, j) can swap, $p_i = p_j$. Moreover, by assumption inequalities (1) and (2) hold.

Consider the case $j = n$. Then $p_i = p_n$ and inequalities (5) and (6) follow, so i is swappable.

Now consider $j < n$. Then

$$\begin{aligned} \text{one}_i^n(p) &= \text{one}_i^j(p) + \text{one}_{j-1}^n(p) \\ &\geq \text{one}_i^j(p) + p_j + p_n \\ &> \overline{\text{one}}_i^j(p) + p_j + p_n \quad \text{by inequality (1)} \\ &= \overline{\text{one}}_i^j(p) + p_i + p_n \\ &\geq \overline{\text{one}}_i^n(p) + p_i + p_n \\ &\geq \overline{\text{one}}_i^n(p) + p_i - p_n \end{aligned}$$

and so inequality (5) holds. Analogously, inequality (6) follows, and thus i is swappable. By symmetry, j is swappable as well. □

Proposition 11. *Let (n, p, q) be as in Definition 2. If $i \in L$ is not swappable, then no $i' \in L$ with $i' \geq i$ can swap. If $j \in R$ is not swappable, then no $j' \in R$ with $j' \leq j$ can swap.*

Proof. By assumption

$$\text{one}_i^n(p) \leq \overline{\text{one}}_i^n(p) + p_i - p_n.$$

It follows that

$$\begin{aligned} \text{one}_{i'}^n(p) &\leq \text{one}_i^n(p) - p_{i'} \\ &\leq \overline{\text{one}}_i^n(p) + p_i - p_n - p_{i'} \\ &\leq \overline{\text{one}}_{i'}^n(p) - p_i + p_i - p_n - p_{i'} \\ &= \overline{\text{one}}_{i'}^n(p) - p_n - p_{i'} \\ &\leq \overline{\text{one}}_{i'}^n(p) - p_n + p_{i'} \end{aligned}$$

and, with an analogous argument for the number of zeros, and by symmetry, the claim follows. $\square$

Proposition 12. *Let (n, p, q) be as in Definition 2. If $i \in L$ is swappable, then all $i' \in L$ with $i' \leq i$ are swappable. If $j \in R$ is swappable, then all $j' \in R$ with $j' \geq j$ are swappable.*

Proof. By assumption

$$\text{one}_i^n(p) > \overline{\text{one}}_i^n(p) + p_i - p_n.$$

It follows that

$$\begin{aligned} \text{one}_{i'}^n(p) &\geq \text{one}_i^n(p) + p_i \\ &> \overline{\text{one}}_i^n(p) + p_i - p_n + p_i \\ &\geq \overline{\text{one}}_{i'}^n(p) + p_{i'} + p_i - p_n + p_i \\ &\geq \overline{\text{one}}_{i'}^n(p) + p_{i'} - p_n \end{aligned}$$

and, with an analogous argument for the number of zeros, and by symmetry, the claim follows. $\square$

Lemma 13. *Let (n, p, q) be as in Definition 2. An index $i \in N$ is swappable for (n, p, q) if and only if it is swappable for every (n, p', q) with $p' \sim p$.*

Proof. We first show: $i \in L$ swappable $\Rightarrow i$ is swappable for every (n, p', q) with $p' \sim p$. It suffices to show that i remains swappable after an arbitrary swap. So consider a swap (i', j) with $i' \neq i$. Since (i', j) can swap, they are also swappable by Proposition 10. Let $p' = \tau_{i'j}(p)$. We distinguish 4 cases:

1. $i' > i$ and $j = n$: By assumption

$$\text{one}_i^n(p') \geq \text{one}_{i'}^n(p') + p_{i'}' \quad \text{since } j = n \quad (9)$$

$$\overline{\text{one}}_i^n(p) \leq \overline{\text{one}}_{i'}^n(p) - p_i \quad (10)$$

$$\text{one}_{i'}^n(p) = \text{one}_{i'}^n(p') \quad (11)$$

$$\overline{\text{one}}_i^n(p) = \overline{\text{one}}_i^n(p') \quad (12)$$

we conclude

$$\begin{aligned}
\text{one}_i^n(p') &\geq \text{one}_{i'}^n(p') + p_{i'}' && \text{inequality (9)} \\
&= \text{one}_{i'}^n(p) + p_{i'}' && \text{equation (11)} \\
&> \overline{\text{one}}_{i'}^n(p) + \underbrace{p_{i'}' + p_{i'} - p_n}_{=1} && \text{by inequality (5)} \\
&= \overline{\text{one}}_{i'}^n(p) + 1 - p_n \\
&\geq \overline{\text{one}}_i^n(p) + p_i + 1 - p_n && \text{equation (10)} \\
&= \overline{\text{one}}_i^n(p') + p_i + 1 - p_n && \text{equation (12)} \\
&= \overline{\text{one}}_i^n(p') + p_i + p_n' \\
&\geq \overline{\text{one}}_i^n(p') + p_i' - p_n'.
\end{aligned}$$

Analogous estimates for the number of zeros give swappability of i for p' .

2. $i' > i$ and $j < n$: Then $p_i' = p_n' = p_i = p_n$. It follows that

$$\begin{aligned}
\text{one}_i^n(p') &\geq \text{one}_{i'}^j(p') + p_{i'}' + p_n' \\
&> \overline{\text{one}}_{i'}^j(p') + p_{i'}' + p_n' && \text{since } (i', j) \text{ can swap in } p' \\
&\geq \overline{\text{one}}_i^n(p') + p_i' + p_n' + p_{i'}' + p_n' \\
&\geq \overline{\text{one}}_i^n(p') + p_i' - p_n'.
\end{aligned}$$

With an analogous estimate for the number of zeros, i remains swappable after the swap.

3. $i' < i$ and $j < n$: Here

$$\begin{aligned}
\text{one}_i^n(p') &= \text{one}_i^n(p) + p_j' - p_j \\
&> \overline{\text{one}}_i^n(p) + p_i - p_n + p_j' - p_j && \text{by inequality (5)} \\
&= \overline{\text{one}}_i^n(p') - p_{i'}' + p_{i'} + p_i - p_n + p_j' - p_j \\
&= \overline{\text{one}}_i^n(p') + p_i - p_n.
\end{aligned}$$

With an analogous estimate for the number of zeros, i thus remains swappable after the swap.

4. It remains to consider the case $i' < i$ and $j = n$. Here

$$\begin{aligned}
\text{one}_i^n(p') &= \text{one}_i^n(p) \\
&> \overline{\text{one}}_i^n(p) + p_i - p_n && \text{by inequality (5)} \\
&= \overline{\text{one}}_i^n(p') - p_{i'}' + p_{i'} + p_i - p_n \\
&= \overline{\text{one}}_i^n(p') - p_{i'}' + p_i \\
&= \overline{\text{one}}_i^n(p') - p_n' + p_i'.
\end{aligned}$$

With analogous reasoning for the number of zeros, i remains swappable.

Overall, swappability is thus preserved. It remains to show that non-swappability is also preserved. So suppose

$$\text{one}_i^n(p) \leq \overline{\text{one}}_i^n(p) + p_i - p_n.$$

1. $i' > i$: This case cannot occur, since by Proposition 11 such an i' is not swappable.

2. $i' < i$ and $j < n$: Here

$$\begin{aligned}
\text{one}_i^n(p') &= \text{one}_i^n(p) + p'_j - p_j \\
&\leq \overline{\text{one}}_i^n(p) + p_i - p_n + p'_j - p_j \\
&= \overline{\text{one}}_i^n(p') - p'_{i'} + p_{i'} + p_i - p_n + p'_j - p_j \\
&= \overline{\text{one}}_i^n(p') + p_i - p_n \\
&= \overline{\text{one}}_i^n(p') + p'_i - p'_n
\end{aligned}$$

and, by the same argument as before, i remains non-swappable.

3. $i' < i$ and $j = n$: Here

$$\begin{aligned}
\text{one}_i^n(p') &= \text{one}_i^n(p) \\
&\leq \overline{\text{one}}_i^n(p) + p_i - p_n \\
&= \overline{\text{one}}_i^n(p') - p'_{i'} + p_{i'} + p_i - p_n \\
&= \overline{\text{one}}_i^n(p') - p'_{i'} + p_i \\
&= \overline{\text{one}}_i^n(p') - p'_n + p'_i
\end{aligned}$$

and again, by the same argument as above, i remains non-swappable.

Overall, it follows that non-swappability of $i \in L$ is preserved under swaps, and by symmetry we can extend this property to $i \in N$. $\square$

Corollary 14. *Given (n, p, q) swappable. Every (n, p', q) with $p' \sim p$ is swappable.*

Definition 15. *Let (n, p, q) be as in Definition 2. We define the left balance $b_L(p)$ and the right balance $b_R(p)$ by*

$$\begin{aligned}
b_L(p) &= \#\{i \in L \mid p_i = 1, q_i = 0\} - \#\{i \in L \mid p_i = 0, q_i = 1\} \\
b_R(p) &= \#\{j \in R \mid p_j = 1, q_j = 0\} - \#\{j \in R \mid p_j = 0, q_j = 1\}
\end{aligned}$$

We call p balanced if

$$b_L(p) = b_R(p).$$

Lemma 16. *Let (n, p, q) be as in Definition 2. For (n, p, q) to be solvable, p must be balanced.*

Proof. By Corollary 6, every swap increases or decreases the left balance and the right balance simultaneously by 1. In particular, the difference between the two balances never changes. Since $b_L(p) = b_R(p) = 0$ trivially holds in the case $p = q$, p must be balanced. $\square$

Definition 17. *Let (n, p, q) be as in Definition 2. We define*

$$\sigma(n, p, q) = \#\{i \in N \mid p_i \neq q_i\}.$$

Lemma 18. *Given (n, p, q) swappable and balanced with $\sigma(n, p, q) > 0$. Then there exists (n, p', q) swappable with $p' \sim p$ and $\sigma(n, p, q) \geq \sigma(n, p', q)$, where swappable $i \in L$ and $j \in R$ exist with $p_i = p_j \neq q_i = q_j$.*

Proof. For $i \in L$ we may assume without loss of generality that $p_i = 0, q_i = 1$. Since the problem is balanced, there is either a $j \in R$ with $p_j = 0, q_j = 1$ or an $i' \in L$ with $p_{i'} = 1, q_{i'} = 0$. In the first case we would already be done. In the second case we use the fact that i and i' are swappable, and can swap one of the two with n . $\square$

Theorem 19. *Let (n, p, q) be as in Definition 2. Then (n, p, q) is solvable if and only if it is center-equal, balanced, and swappable.*

Proof. That a solvable (n, p, q) must be center-equal follows from Corollary 8. That a solvable p must also be balanced follows from Lemma 16. That a solvable p also implies that (n, p, q) is swappable is clear from Corollary 14 as well. For the converse direction we assume that (n, p, q) is center-equal, balanced, and swappable, and suppose the problem were nevertheless not solvable. Then we can keep performing swaps until we reach a smallest number of unsatisfiable swaps. Consider the corresponding $\bar{p} \sim p$. Then $\sigma(n, \bar{p}, q) > 0$. Moreover, since $\bar{p} \sim p$, $(n, \bar{p}, q)$ is also swappable by Corollary 14. Since (n, p, q) is center-equal, $p_{\frac{n}{2}} = q_{\frac{n}{2}}$ and $p_{\frac{n}{2}+1} = q_{\frac{n}{2}+1}$ hold, and by Corollary 8 applied to $\bar{p} \sim p$, we also have $\bar{p}_{\frac{n}{2}} = q_{\frac{n}{2}}$ and $\bar{p}_{\frac{n}{2}+1} = q_{\frac{n}{2}+1}$. Hence $\sigma(n, \bar{p}, q) > 0$ can only be caused by an index in $L \cup R$, i.e. there is $i \in L$ with $\bar{p}_i \neq q_i$ or $j \in R$ with $\bar{p}_j \neq q_j$ – exactly the hypothesis required by Lemma 18. By Lemma 18 we may therefore assume that $i \in L$ and $j \in R$ exist with $p_i = p_j \neq q_i = q_j$. Now there are two cases:

1. $\bar{p}_1 = \bar{p}_n$: If $\bar{p}_i = \bar{p}_j = \bar{p}_1 = \bar{p}_n$, we set $p' = (\tau_{in} \circ \tau_{1j})(\bar{p})$, otherwise $p' = (\tau_{in} \circ \tau_{1j} \circ \tau_{1n})(\bar{p})$.
2. $\bar{p}_1 \neq \bar{p}_n$: Here we assume without loss of generality that $\bar{p}_i = \bar{p}_n$. We then set $p' = (\tau_{1j} \circ \tau_{1n} \circ \tau_{in})(\bar{p})$.

In every case, by performing further swaps we obtain (n, p', q) with $\sigma(n, p', q) < \sigma(n, \bar{p}, q)$, a contradiction. $\square$

Corollary 20. *It can be determined in $O(n)$ time whether the problem is solvable.*

Proof. By Theorem 19 we only need to check whether (n, p, q) is center-equal, balanced, and swappable. Whether (n, p, q) is center-equal can be checked directly from the two values $p_{\frac{n}{2}}, q_{\frac{n}{2}}$ and $p_{\frac{n}{2}+1}, q_{\frac{n}{2}+1}$ in $O(1)$. The balances can be computed in $O(n)$. Afterwards, by Proposition 12, one only needs to check whether $\max_{i \in L, p_i \neq q_i} i$ is swappable with n and whether $\min_{j \in R, p_j \neq q_j} j$ is swappable with 1. Each of these can be done in $O(n)$ time with $O(1)$ memory. $\square$